

RoboWM-Bench: A Benchmark for Evaluating World Models in Robotic Manipulation

Abstract—Recent advances in large-scale video world models have enabled increasingly realistic future prediction, raising the prospect of using generated videos as scalable supervision for robot learning. However, for embodied manipulation, perceptual realism alone is not sufficient: generated interactions must also be physically consistent and executable by robotic agents. Existing benchmarks provide valuable assessments of visual quality and physical plausibility, but they do not systematically evaluate whether predicted behaviors can be translated into executable actions that complete manipulation tasks. We introduce RoboWM-Bench, a manipulation-centric benchmark for embodiment-grounded evaluation of video world models. RoboWM-Bench converts generated human-hand and robotic manipulation videos into embodied action sequences and validates them through execution in physically grounded simulation environments. Built on real-to-sim scene reconstruction and diverse manipulation tasks, RoboWM-Bench enables standardized, reproducible, and scalable evaluation of physical executability. Using RoboWM-Bench, we evaluate state-of-the-art video world models and observe that visual plausibility and embodied executability are not always aligned. Our analysis highlights several recurring factors that affect execution performance, including spatial reasoning, contact prediction, and non-physical geometric distortions, particularly in complex and long-horizon interactions. These findings provide a more fine-grained view of current model capabilities and underscore the value of embodiment-aware evaluation for guiding physically grounded world modeling in robotic manipulation. The project webpage is available at <https://sites.google.com/view/robowm-bench>.

I. INTRODUCTION

Recent advances in large-scale video generation have produced increasingly realistic video world models capable of synthesizing temporally coherent and visually plausible futures [1]–[4]. While these models create new opportunities for robot learning from predicted videos, embodied manipulation requires more than visual fidelity: generated interactions must respect physical constraints and support executable control. Accordingly, recent work has explored robotics-oriented video world models that better capture robot–environment interactions and embodiment-aware dynamics [5]–[10]. As predicted manipulation videos become a promising source of scalable supervision for robot learning [6], [11], [12], reliable evaluation is essential for determining whether they are physically grounded and useful for downstream control.

Existing benchmarks for video world models have provided valuable tools for evaluating visual fidelity, semantic consistency, and temporal coherence [13]–[18]. More recent efforts further extend evaluation toward physical plausibility, offering useful diagnostics of whether generated videos preserve coherent dynamics [19]–[21]. For robotic manipulation, however, physical plausibility raises a more operational

question: can the behaviors depicted in predicted videos be translated into executable actions that accomplish the intended task? While recent work [22] takes an important step toward embodiment-grounded evaluation through real-world robot validation, broad and reproducible assessment remains challenging due to the cost and limited scalability of physical experiments.

To address this gap, we introduce **RoboWM-Bench**, a manipulation-centric benchmark for evaluating video world models through execution-based validation in high-fidelity simulation environments. RoboWM-Bench covers both human-hand and robotic manipulation scenarios, reflecting complementary settings for learning from human demonstrations and robot-centric control. As illustrated in Figure 1, for each task, a video world model is conditioned on an initial observation and task description to generate a future manipulation video. The predicted behaviors are then converted into executable action sequences through inverse dynamics modeling [11], [23] for robotic videos, or pose tracking and retargeting [24]–[26] for human demonstrations. These actions are executed in high-fidelity real-to-simulation environments, enabling standardized and reproducible validation of physical executability. RoboWM-Bench combines step-level executability checks with final task-level success rates, supporting both fine-grained diagnosis and holistic assessment. Spanning diverse object dynamics, task horizons, and single-arm and bimanual interactions, RoboWM-Bench provides an accessible and scalable protocol for comparing video world models under embodiment constraints.

We conduct extensive experiments with RoboWM-Bench to evaluate state-of-the-art video world models under embodied execution. The results suggest that visual realism does not always translate into physical executability, with execution success tending to decrease as task complexity increases, especially in long-horizon interactions and deformable-object manipulation. Qualitative analysis identifies recurring sources of execution difficulty, including inaccurate contact prediction, unrealistic object deformation, and geometric distortions that can lead to dynamically infeasible actions. Although fine-tuning on manipulation-specific data improves executability, generating physically consistent interactions remains challenging. We further validate the evaluation pipeline through action-extraction accuracy and real-to-sim execution consistency analyses, supporting the reliability of these findings. Together, these results demonstrate the value of RoboWM-Bench as a diagnostic protocol for embodiment-grounded evaluation and highlight opportunities for advancing physically grounded world models for robotic

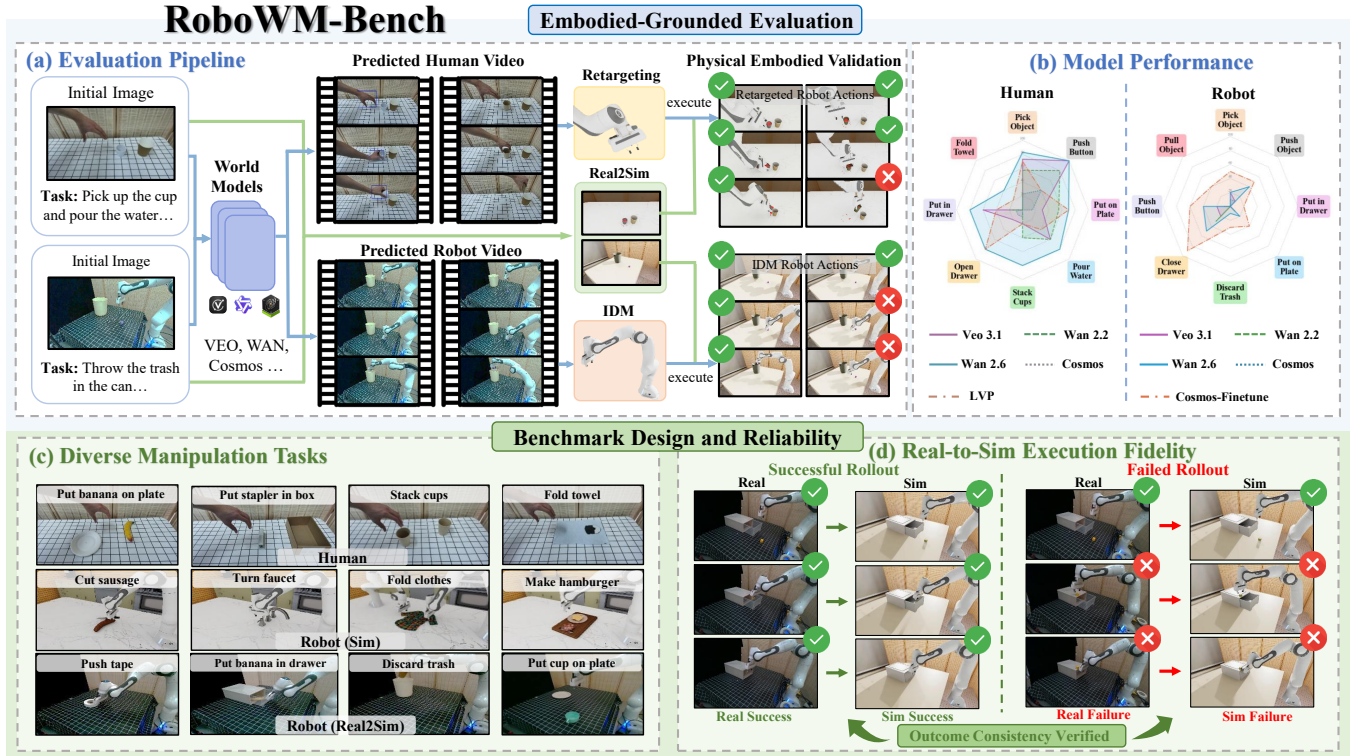


Fig. 1. **Overview of RoboWM-Bench.** RoboWM-Bench evaluates video world models through embodiment-grounded execution. (a) Predicted human-hand and robotic manipulation videos are converted into executable actions via retargeting or inverse dynamics model (IDM) and validated in real-to-sim simulation environments. (b) The benchmark compares execution performance across state-of-the-art video world models for human-hand and robotic manipulation. (c) RoboWM-Bench covers diverse tasks spanning different object properties, interaction dynamics, temporal horizons, and coordination requirements. (d) Real-to-sim fidelity is evaluated by replaying identical real-world trajectories in reconstructed simulations and comparing the resulting task outcomes.

manipulation.

II. RELATED WORK

A. World Models for Robotics

Recent advances in large-scale video generation have renewed interest in world models as predictive models of physical dynamics. Models such as Sora [1], Veo [2], Wan [3], and Cosmos [5], [10] demonstrate strong visual realism and temporal coherence. For robotic manipulation, however, generated videos must also preserve the physical and interaction dynamics required for executable control. This has motivated robotics-oriented world models for video-based planning [6], scalable robot-data generation [9], [11], embodied interaction prediction [7], [8], and compositional video planning [27]. As these models increasingly serve as simulators, planners, or data engines, their evaluation must extend beyond visual fidelity to physical consistency and control feasibility.

B. Learning Robotic Actions from Video

Prior work bridges videos and robotic actions through several complementary approaches. Representation-learning methods such as R3M [28] and VIP [29] pretrain visual encoders on diverse videos for downstream control. Latent-action and domain-adaptation methods infer action-relevant structure without dense robot annotations [30], [31], while inverse-dynamics approaches recover executable actions from observed or generated videos [11], [32]. Human-

to-robot transfer methods retarget in-the-wild human videos to robot policies, as in Phantom [25] and Masquerade [26], and recent VLA models show emergent transfer through multimodal pretraining [33]. In this work, these video-to-action interfaces are used as evaluation tools for testing whether generated manipulation videos support successful embodied execution.

C. Evaluation and Benchmarks of World Models

Existing video-generation benchmarks primarily assess perceptual realism, temporal consistency, and text-video alignment [13]–[17]. More recent benchmarks extend evaluation toward physical reasoning and embodied-world-model capabilities [18]–[21], but action-level task completion under robotic constraints remains less directly measured. Robotics-oriented studies have begun to incorporate execution-based validation: LVP [6] evaluates video-conditioned planning on real-world manipulation tasks, while Wow-wo-val [22] introduces an embodied evaluation protocol based on real-robot validation. Complementary to these efforts, RoboWM-Bench provides standardized task suites and a unified execution-based protocol across simulation-native and real-to-sim reconstructed scenarios, enabling scalable assessment of physical executability and action-level consistency.

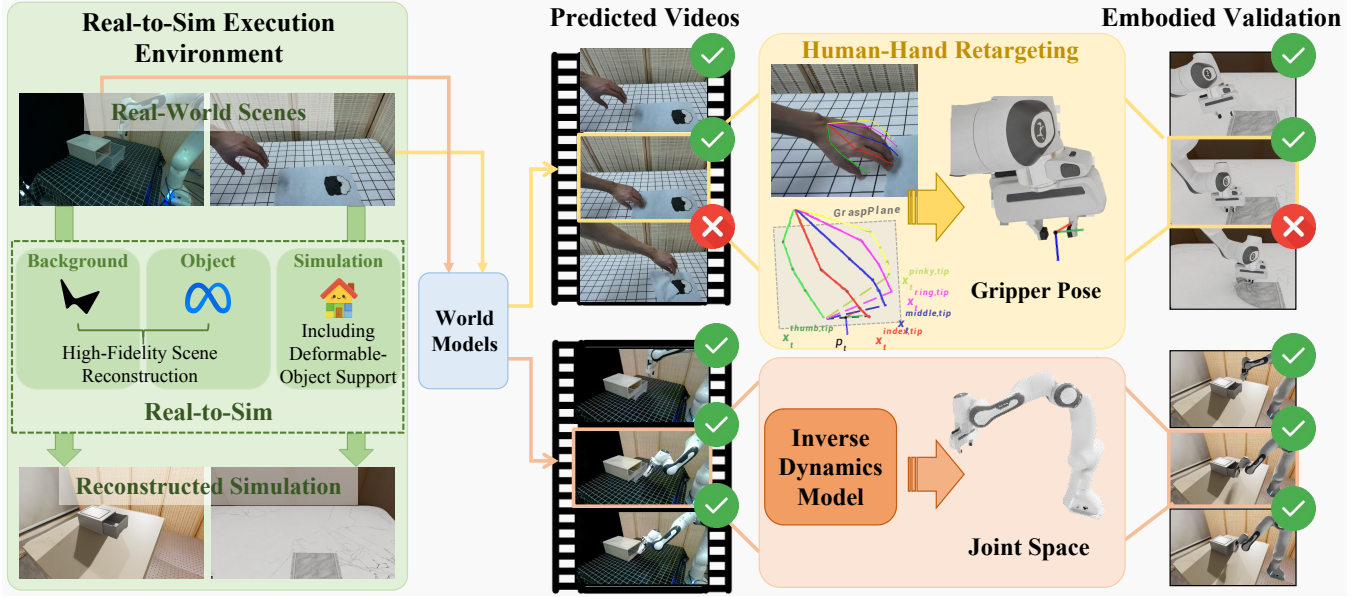


Fig. 2. **Pipeline of RoboWM-Bench.** Given an initial scene observation, the corresponding real-world scene is reconstructed in simulation through a real-to-sim pipeline, enabling consistent and reproducible evaluation. Predicted videos are then converted into executable robot actions through two pathways: human-centric retargeting, which estimates 3D hand poses and retargets them to robot end-effector actions, and robot-centric inverse dynamics, which predicts joint-space actions via an inverse dynamics model (IDM). The resulting actions are executed in simulation and evaluated using step-level checkers and final task success rates to measure embodied executability.

III. ROBOWM-BENCH

A. Benchmark Overview

We introduce RoboWM-Bench, a benchmark for evaluating video world models through embodiment-grounded validation, as illustrated in Figure 2. RoboWM-Bench treats physical executability as a measurable criterion by testing whether behaviors inferred from generated videos can complete manipulation tasks when executed. First, RoboWM-Bench provides standardized simulation environments for reproducible evaluation, covering both simulation-native scenarios and real-to-sim reconstructions of real-world scenes (Section III-B). Given an initial scene observation and a task description, a video world model predicts a future manipulation video in either a human-hand or robotic setting. The benchmark then provides video-to-action interfaces that convert the generated videos into action sequences, which are executed in the corresponding simulation environments (Section III-C). Finally, RoboWM-Bench defines a manipulation task suite spanning different object properties, interaction regimes, and temporal horizons (Section III-D), and evaluates executability through step-level verification and task-level success (Section III-E).

B. High-Fidelity Real-to-Sim Framework

To support accessible, reproducible, and scalable evaluation, RoboWM-Bench conducts embodiment-grounded validation in open-source simulation environments. It covers both simulation-native scenarios and real-to-sim reconstructions of real-world scenes, enabling predicted interactions to be evaluated under controlled physical dynamics while reducing dependence on specific physical setups.

To support high-fidelity execution, RoboWM-Bench builds on the LeHome simulation engine [34], which supports

household manipulation scenarios involving rigid, articulated, and deformable objects. We design simulation-native scenarios using LeHome engine, and reconstruct real-world scenes with a modular real-to-sim pipeline covering scene reconstruction, object modeling, and pose calibration. Inspired by recent work [35], background scenes are reconstructed using 4D Gaussian representations to preserve visual realism and spatial consistency. For interactive objects, rigid geometries are obtained through 3D segmentation and reconstruction [36], while articulated and deformable object pairs are constructed following [34]. Object poses are estimated using pose estimation models [37], [38], and camera poses are calibrated with FEEPE [39], averaging results across multiple runs for stability. We further validate the fidelity of the reconstructed environments in Section IV-D. This modular design preserves the physical structure and spatial configuration of real scenes and supports new evaluation scenarios through either simulation-native assets or reconstructed real-world scenes.

C. Embodied Video-to-Action Execution

To evaluate generated videos through execution, RoboWM-Bench converts predicted manipulation videos into executable action sequences. The benchmark supports both human-hand and robotic videos, covering complementary evaluation settings: human-hand videos reflect interaction patterns that current world models often generate more reliably, while robotic videos are more directly aligned with downstream robot policy learning and control. We therefore use separate video-to-action pathways for the two settings.

1) *Human-Centric Retargeting:* Inspired by prior work [25], [26], we estimate human hand poses from videos and retarget them to robot end-effector actions. We

reconstruct 3D hand poses using HaMeR [24] and use the recovered keypoints to derive three control components: end-effector position, orientation, and gripper opening.

The gripper target position is defined as the midpoint between the thumb and index fingertips. For orientation, instead of relying on global finger configurations as in prior work [25], we use contact-relevant geometry: we fit a plane through the thumb and index finger keypoints, project the two fingertips onto this plane, and define the x -axis as the line connecting the projections, with the z -axis given by the plane normal. This yields more stable end-effector poses while preserving local human-object interaction geometry. For gripper opening, we use the minimum distance between the thumb tip and all other fingertips rather than the thumb-index distance, accounting for cases where the index fingertip is not the primary contact point. We then apply trajectory smoothing and temporal denoising to stabilize the retargeted motion signals [25], [26].

2) *Robot-Centric Execution*: For robotic manipulation videos, we recover action sequences using an inverse dynamics model (IDM), following established video-to-action formulations in prior work [11], [23]. We adopt the IDM architecture from [11], which takes two consecutive image frames as input and predicts the intermediate joint-space action chunk. We train the IDM with a two-stage strategy: large-scale simulation data provides action-labeled supervision for motion pretraining, while a small set of real-world trajectories is used for visual adaptation.

Specifically, we first collect simulation trajectories with a Franka arm in a physics simulator, recording paired RGB observations and joint-space actions. Compared with real-world collection, simulation enables action-labeled trajectories to be generated more efficiently and at higher temporal resolution, providing dense and smooth supervision for inverse-dynamics learning. To keep simulation pretraining scalable and focused on robot motion, we apply background masking, inspired by [9], retaining only the robot arm in simulation videos. We then finetune the IDM on real-world trajectories collected with a physical Franka arm, without background masking. This design leverages scalable simulation supervision while adapting to real observations, enabling reliable action extraction from generated robotic videos for embodiment-grounded evaluation.

D. Manipulation Task Suite with Diverse Complexity

RoboWM-Bench includes a diverse suite of manipulation tasks designed to evaluate the embodied reasoning and physical consistency of video world models across varying levels of complexity. The tasks span different object properties, interaction dynamics, temporal horizons, and coordination requirements, enabling systematic assessment beyond short-horizon rigid-object manipulation.

The task suite covers several representative interaction regimes. Basic rigid-object tasks, such as object pickup and trash disposal, primarily assess contact precision and spatial reasoning. Articulated-object tasks, including drawer opening and faucet rotation, require models to capture kinematic

constraints and structured motion. Deformable-object tasks, such as towel folding, further evaluate whether models can generate physically plausible non-rigid interactions.

RoboWM-Bench also includes more complex manipulation settings that require longer-term reasoning or bimanual coordination. Long-horizon compositional tasks, such as assembling a hamburger, test multi-stage planning and temporal consistency, while bimanual tasks, such as object handover and collaborative towel folding, introduce coordination constraints between two hands. Together, this structured task design supports fine-grained evaluation of how video world models handle increasingly complex embodied interactions.

E. Evaluation of Embodied Executability

We define *embodied executability* as whether behaviors inferred from generated videos can be executed to complete the intended task. RoboWM-Bench evaluates executability at two levels: step-level verification and task-level success. For each task, we predefine a set of key action nodes corresponding to semantically meaningful interaction stages, such as contact events (e.g., grasping) or stable end-effector configurations (e.g., lifting). During execution, step-level verification checks whether the predicted behavior satisfies the required interaction and dynamical constraints at each key node. A trajectory is considered task-level successful only if all key nodes pass these checks and the final task objective is achieved. This hierarchical protocol supports fine-grained failure diagnosis while providing a clear and measurable criterion for overall task completion.

IV. EXPERIMENTS

We evaluate state-of-the-art video world models on RoboWM-Bench across a diverse set of human-hand and robotic manipulation tasks (Section IV-A). Our experiments first quantify embodied executability and characterize common failure modes (Section IV-B). We then compare RoboWM-Bench with PAI-Bench on the same generated videos to examine the complementary role of execution-based evaluation (Section IV-C). Finally, we validate the reliability of RoboWM-Bench through action-extraction and simulation-consistency analyses (Section IV-D).

A. Environment Setup

1) *Tasks and Environments*: We evaluate video world models on the RoboWM-Bench task suite, which covers both human-hand and robotic manipulation tasks. In each robotic task setup, a Franka arm operates on a tabletop where objects are initialized with randomized poses, and the scene is observed from a camera viewpoint that remains fixed during evaluation. We consider simulation-native scenarios and real-world scenes reconstructed through our real-to-sim pipeline, enabling generated robotic behaviors to be evaluated under a unified simulation-based execution protocol.

For human-hand tasks, we use real-world scenes with randomized object poses. Following the input convention of LVP [6], the initial observation includes a visible human hand positioned above the tabletop, ensuring fair comparison

TABLE I

EMBODIED EXECUTION SUCCESS RATES (%) ON ROBOWM-BENCH FOR HUMAN-HAND (TOP TWO SECTIONS) AND ROBOTIC MANIPULATION TASKS (BOTTOM TWO SECTIONS), REPORTED AT BOTH TASK AND STEP LEVELS.

Human (Task Level)								
Method	Pick Object	Push Button	Put on Plate	Pour Water	Stack Cups	Open Drawer	Put in Drawer	Fold Towel
Cosmos	23%	40%	15%	0%	10%	10%	10%	0%
Wan 2.2	57%	80%	55%	60%	40%	0%	20%	0%
Veo 3.1	73%	100%	30%	60%	20%	20%	60%	0%
Wan 2.6	83%	100%	70%	80%	80%	80%	80%	40%
LVP	70%	40%	70%	40%	20%	80%	40%	20%

Human (Step Level)								
Method	contact	Put on Plate lift	place	contact	lift	Put in Drawer above drawer	in drawer	close drawer
Cosmos	90%	20%	15%	80%	20%	20%	20%	10%
Wan 2.2	100%	60%	55%	100%	60%	60%	40%	20%
Veo 3.1	100%	70%	30%	100%	70%	70%	60%	60%
Wan 2.6	100%	75%	70%	100%	80%	80%	80%	80%
LVP	100%	75%	70%	100%	70%	60%	50%	40%

Robot (Task Level)								
Method	Close Drawer	Pick Object	Push Object	Push Button	Put on Plate	Discard Trash	Pull Object	Put in Drawer
Cosmos	0%	10%	10%	10%	10%	0%	0%	0%
Wan 2.2	30%	10%	0%	0%	0%	0%	0%	0%
Veo 3.1	20%	20%	10%	20%	10%	0%	0%	0%
Wan 2.6	50%	20%	40%	40%	20%	10%	0%	0%
Cosmos-FT	90%	50%	50%	60%	40%	30%	40%	20%

Robot (Step Level)								
Method	contact	Put on Plate lift	place	contact	lift	Put in Drawer above drawer	in drawer	close drawer
Cosmos	30%	10%	10%	10%	0%	0%	0%	0%
Wan 2.2	20%	0%	0%	0%	0%	0%	0%	0%
Veo 3.1	40%	10%	10%	30%	0%	0%	0%	0%
Wan 2.6	40%	20%	20%	30%	0%	0%	0%	0%
Cosmos-FT	60%	40%	40%	60%	20%	20%	20%	20%

across baselines. We focus on real-world scenes because simulating a real human hand within the physics engine would not faithfully capture human manipulation dynamics and would therefore provide limited evaluation value.

All executions are conducted in the LeHome simulation environment [34], [40]. To ensure fair and reproducible comparison, each task uses 10 different initial object configurations shared across all models, together with standardized task descriptions, random seeds, and evaluation protocols. We report both task- and step-level success rates.

2) *Baselines*: We evaluate state-of-the-art video world models covering both general-purpose and interaction-oriented generation. For general-purpose video generation, we include closed-source models, Veo3.1 [2] and Wan2.6 [3], as well as open-source models, Wan2.2 [3] and Cosmos-Predict2.5 [5]. For interaction-oriented generation, we include LVP [6], which is specifically trained to capture complex human interactive behaviors. To further investigate the potential of video world models for embodied manipulation, we fine-tune Cosmos-Predict2.5 on our collected real-world manipulation dataset with 50 trajectories per task and denote the resulting variant as Cosmos-Finetune.

B. Embodied Executability of Video World Models

Table I reports task- and step-level execution success rates. Results on simulation-native robotic tasks are provided on the webpage due to space limitations. The experimental results reveal several consistent trends, which we analyze below.

First, **human-hand videos achieve higher execution success than robotic manipulation videos**. This gap likely

reflects biases in large-scale video pretraining, where human interactions are far more prevalent than robotic data. Generated human hands preserve more stable geometry during interaction, whereas robotic manipulators are more likely to exhibit structural distortions that lead to execution failures.

Second, **execution success decreases as task complexity increases**. Short-horizon tasks such as *Push Button* are generally easier, while longer-horizon tasks such as *Put in Drawer* suffer from error accumulation across multiple interaction stages. Among human-hand tasks, *Fold Towel* remains particularly challenging, reflecting the added complexity of deformable-object interactions.

Third, **manipulation-specific fine-tuning improves robotic executability**. Cosmos-Finetune substantially outperforms its pretrained counterpart, indicating that even limited task-specific data, with 50 trajectories per task, can improve the generation of executable robotic behaviors. Fine-tuning reduces robot-geometry artifacts and improves joint articulation, although errors in 3D object localization and grasp stability remain common failure factors.

Figure 3 presents representative qualitative results from Wan2.6, the strongest overall model in our evaluation. Its success and failure cases therefore provide useful insight into the remaining gap between visual plausibility and embodied executability. These examples show that generated interactions can appear plausible in video while still failing under physical execution. In *Put on Plate*, for instance, the predicted video shows the fingers merely touching the object without forming a stable grasp, yet the object is lifted. Such interactions are physically implausible and fail during

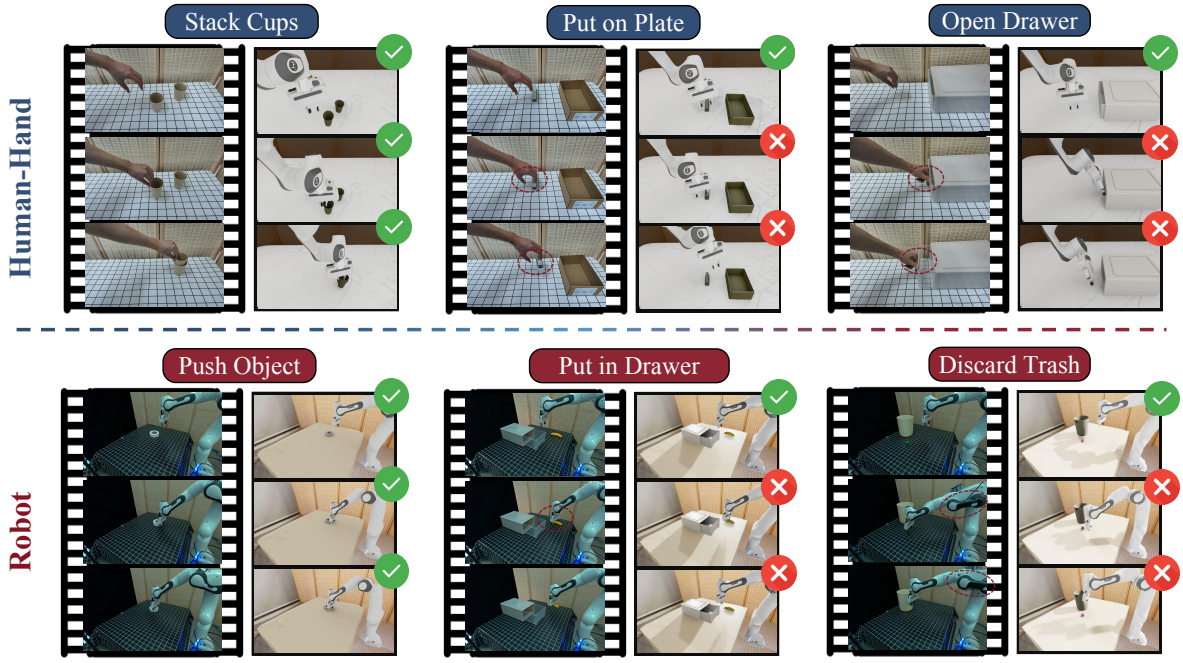


Fig. 3. Execution results on RoboWM-Bench. For each task, predicted videos (left) are converted to robot actions and executed in simulation (right).

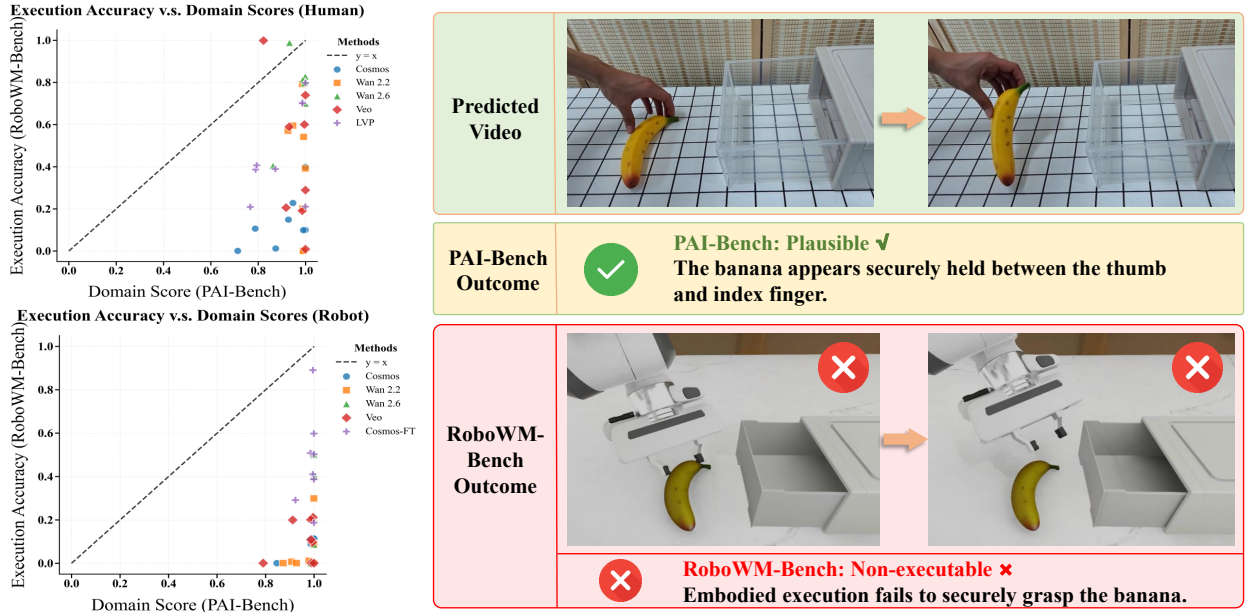


Fig. 4. Comparison between PAI-Bench and RoboWM-Bench.

execution. In *Open Drawer*, the predicted motion resembles closing the drawer and does not establish a proper grasp, which is reflected by the simulated execution outcome. For robotic manipulation videos, failures can also arise from geometric distortions of the generated robot arm. The end effector may appear to reach the target in the video, while the recovered joint configuration yields a mismatched end-effector pose under the robot’s true forward kinematics.

C. Perceptual Plausibility vs. Embodied Executability

We compare execution accuracy in RoboWM-Bench with the domain scores in PAI-Bench, a commonly used metric for evaluating the perceptual plausibility of generated videos.

As shown in Figure 4, the same predicted videos obtain near-saturated scores on PAI-Bench across different world models, whereas RoboWM-Bench evaluates whether the predicted behaviors are physically executable, resulting in more discriminative outcomes. This discrepancy arises because some actions may appear visually plausible (Figure 3) yet remain physically infeasible, an issue that perceptual domain scores may not capture but becomes evident under embodied execution. These results highlight the complementary role of RoboWM-Bench and show that embodiment-grounded evaluation provides a more direct measure of physical executability. Further PAI-Bench analysis and domain-score

implementation details are provided on the project webpage.

D. Reliability Analysis of RoboWM-Bench

To examine the reliability of RoboWM-Bench, we validate two key interfaces in the evaluation pipeline: real-to-sim execution consistency and video-to-action replay success. The former tests whether reconstructed simulation environments preserve real-world task outcomes, while the latter evaluates whether retargeting or inverse dynamics can recover executable actions from videos.

1) *Real-to-Sim Execution Consistency*: We first assess whether reconstructed simulation environments preserve real-world execution outcomes. Specifically, we execute real-world robotic trajectories in the corresponding reconstructed simulation environments and compare task outcomes across domains. We use robotic trajectories because they provide the same low-level action sequences in both real and simulated scenes, keeping this test focused on reconstruction fidelity; video-to-action conversion, including human-to-robot retargeting, is evaluated separately in the next subsection.

For each task, we collect 10 successful and 10 failed real-world trajectories and execute the same actions in the reconstructed simulation environments. Figure 5 shows that simulation reproduces success and failure outcomes. This consistency suggests that the reconstructed environments preserve the task-relevant geometry and interaction constraints needed for execution-based evaluation. Real-world and simulated videos are available on the project webpage.

2) *Video-to-Action Replay Success*: We next evaluate whether the video-to-action modules can recover executable actions from videos. We process videos from successful real-world manipulation trajectories using our action extraction pipeline, and execute the extracted actions in the reconstructed simulation environments to test whether they reproduce the original outcomes.

As shown in Table II, the human-hand pose tracking and retargeting pipeline achieves near-perfect replay success. The remaining failures in *Stack Cups* and *Pour Water* occur in deliberately contact-sensitive cases involving cylindrical objects, where small discrepancies between human fingertip contacts and retargeted robot gripper contacts can destabilize grasps. These cases provide conservative stress tests for retargeting accuracy rather than typical failures observed across most generated videos.

For robotic videos, we compare IDM_{Real} , trained on 50 real-world trajectories per task, with $IDM_{\text{Sim+Real}}$, which uses large-scale simulation pretraining followed by finetuning on the same real-world trajectories. $IDM_{\text{Sim+Real}}$ achieves substantially higher replay success, indicating that scalable action-labeled simulation data provides useful motion priors and improves inverse-dynamics prediction. The remaining failures mainly arise in near-boundary contact cases, such as intentionally shallow grasps, where minor prediction errors can compromise grasp stability.

V. CONCLUSION

We introduced RoboWM-Bench, a manipulation-centric benchmark for evaluating video world models through

embodiment-grounded execution. RoboWM-Bench converts predicted human-hand and robotic manipulation videos into executable actions and validates them in standardized simulation and real-to-sim environments using step-level and task-level execution metrics. Covering diverse object dynamics, task horizons, and coordination requirements, the benchmark provides a scalable and reproducible protocol that complements perceptual video evaluation with execution-based assessment. Experiments with state-of-the-art video world models show that embodied execution reveals additional challenges in complex manipulation settings, particularly those requiring precise contact, deformable-object reasoning, or long-horizon consistency. Overall, RoboWM-Bench offers an execution-grounded diagnostic framework for advancing physically consistent world models for robotic manipulation.

REFERENCES

- [1] T. Brooks, W. Peebles, C. Holmes, W. DePue, Y. Guo, L. Jing, D. Schnurr, J. Taylor, T. Luhman, C. Ng, R. Wang, A. Ramesh, *et al.*, “Video generation models as world simulators,” <https://openai.com/research/video-generation-models-as-world-simulators>, 2024, openAI Technical Report.
- [2] Google DeepMind, “Veo: a text-to-video generation system (veo-3 technical report),” Google DeepMind, Tech. Rep. Veo-3-Tech-Report, 2025, technical Report. [Online]. Available: <https://storage.googleapis.com/deepmind-media/veo/Veo-3-Tech-Report.pdf>
- [3] T. Wan, A. Wang, B. Ai, B. Wen, C. Mao, C.-W. Xie, D. Chen, F. Yu, H. Zhao, J. Yang, *et al.*, “Wan: Open and advanced large-scale video generative models,” *arXiv preprint arXiv:2503.20314*, 2025.
- [4] Y. Gao, H. Guo, T. Hoang, W. Huang, L. Jiang, F. Kong, H. Li, J. Li, L. Li, X. Li, *et al.*, “Seedance 1.0: Exploring the boundaries of video generation models,” *arXiv preprint arXiv:2506.09113*, 2025.
- [5] A. Ali, J. Bai, M. Bala, Y. Balaji, A. Blakeman, T. Cai, J. Cao, T. Cao, E. Cha, Y.-W. Chao, *et al.*, “World simulation with video foundation models for physical ai,” *arXiv preprint arXiv:2511.00062*, 2025.
- [6] B. Chen, T. Zhang, H. Geng, K. Song, C. Zhang, P. Li, W. T. Freeman, J. Malik, P. Abbeel, R. Tedrake, *et al.*, “Large video planner enables generalizable robot control,” *arXiv preprint arXiv:2512.15840*, 2025.
- [7] X. Chi, P. Jia, C.-K. Fan, X. Ju, W. Mi, K. Zhang, Z. Qin, W. Tian, K. Ge, H. Li, *et al.*, “Wow: Towards a world omniscient world model through embodied interaction,” *arXiv preprint arXiv:2509.22642*, 2025.
- [8] S. Huang, L. Chen, P. Zhou, S. Chen, Z. Jiang, Y. Hu, Y. Liao, P. Gao, H. Li, M. Yao, *et al.*, “Enerverse: Envisioning embodied future space for robotics manipulation (2025),” *arXiv preprint arXiv:2501.01895*, 2025.
- [9] G. Team, A. Ye, B. Wang, C. Ni, G. Huang, G. Zhao, H. Li, J. Zhu, K. Li, M. Xu, *et al.*, “Gigaworld-0: World models as data engine to empower embodied ai,” *arXiv preprint arXiv:2511.19861*, 2025.
- [10] N. Agarwal, A. Ali, M. Bala, Y. Balaji, E. Barker, T. Cai, P. Chattopadhyay, Y. Chen, Y. Cui, Y. Ding, *et al.*, “Cosmos world foundation model platform for physical ai,” *arXiv preprint arXiv:2501.03575*, 2025.
- [11] J. Jang, S. Ye, Z. Lin, J. Xiang, J. Bjorck, Y. Fang, F. Hu, S. Huang, K. Kundalia, Y.-C. Lin, *et al.*, “Dreamgen: Unlocking generalization in robot learning through video world models,” *arXiv preprint arXiv:2505.12705*, 2025.
- [12] G. Team, A. Ye, B. Wang, C. Ni, G. Huang, G. Zhao, H. Li, J. Li, J. Zhu, L. Feng, *et al.*, “Gigabrain-0: A world model-powered vision-language-action model,” *arXiv preprint arXiv:2510.19430*, 2025.
- [13] Z. Huang, Y. He, J. Yu, F. Zhang, C. Si, Y. Jiang, Y. Zhang, T. Wu, Q. Jin, N. Chanpaisit, *et al.*, “Vbench: Comprehensive benchmark suite for video generative models,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 21 807–21 818.
- [14] W. Feng, J. Li, M. Saxon, T.-j. Fu, W. Chen, and W. Y. Wang, “Tc-bench: Benchmarking temporal compositionality in text-to-video and image-to-video generation,” *arXiv preprint arXiv:2406.08656*, 2024.



Task	Success Consistency	Failure Consistency
Pick Object	10/10	10/10
Pull Object	10/10	10/10
Push Object	10/10	10/10
Put on Plate	10/10	10/10
Discard Trash	10/10	10/10
Close Drawer	10/10	10/10
Put in Drawer	10/10	10/10
Average	100%	100%

Fig. 5. **Real-to-sim consistency evaluation.** Left: qualitative examples of replaying identical real-world trajectories in reconstructed simulation environments. Right: outcome agreement between real-world and reconstructed simulation executions.

TABLE II
ACCURACY OF ACTION EXTRACTION METHODS.

Method	Human							
	Pick Object	Stack Cups	Pour Water	Open Drawer	Fold Towel	Put on Plate	Put in Drawer	Average
Retargeting	100%	90%	90%	100%	100%	100%	100%	97.1%

Method	Robot							
	Pick Object	Pull Object	Push Object	Discard Trash	Close Drawer	Put on Plate	Put in Drawer	Average
IDM _{Real}	70%	70%	80%	70%	90%	70%	50%	71.4%
IDM _{Sim+Real}	100%	90%	100%	90%	100%	100%	90%	95.7%

- [15] X. Ling, C. Zhu, M. Wu, H. Li, X. Feng, C. Yang, A. Hao, J. Zhu, J. Wu, and X. Chu, "Vmbench: A benchmark for perception-aligned video motion generation," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025, pp. 13 087–13 098.
- [16] P. Ji, C. Xiao, H. Tai, and M. Huo, "T2vbench: Benchmarking temporal dynamics for text-to-video generation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 5325–5335.
- [17] Y. Liu, X. Cun, X. Liu, X. Wang, Y. Zhang, H. Chen, Y. Liu, T. Zeng, R. Chan, and Y. Shan, "Evalcrafter: Benchmarking and evaluating large video generation models," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2024, pp. 22 139–22 149.
- [18] H. Yue, S. Huang, Y. Liao, S. Chen, P. Zhou, L. Chen, M. Yao, and G. Ren, "Ewmbench: Evaluating scene, motion, and semantic quality in embodied world models," *arXiv preprint arXiv:2505.09694*, 2025.
- [19] D. Zheng, Z. Huang, H. Liu, K. Zou, Y. He, F. Zhang, L. Gu, Y. Zhang, J. He, W.-S. Zheng, *et al.*, "Vbench-2.0: Advancing video generation benchmark suite for intrinsic faithfulness," *arXiv preprint arXiv:2503.21755*, 2025.
- [20] F. Zhou, J. Huang, J. Li, D. Ramanan, and H. Shi, "Pai-bench: A comprehensive benchmark for physical ai," *arXiv preprint arXiv:2512.01989*, 2025.
- [21] Y. Shang, Z. Li, Y. Ma, W. Su, X. Jin, Z. Wang, L. Jin, X. Zhang, Y. Tang, H. Su, *et al.*, "Worldarena: A unified benchmark for evaluating perception and functional utility of embodied world models," *arXiv preprint arXiv:2602.08971*, 2026.
- [22] C.-K. Fan, X. Chi, X. Ju, H. Li, Y. Bao, Y.-K. Wang, L. Chen, Z. Jiang, K. Ge, Y. Li, *et al.*, "Wow, wo, val! a comprehensive embodied world model evaluation turing test," *arXiv preprint arXiv:2601.04137*, 2026.
- [23] B. Baker, I. Akkaya, P. Zhokov, J. Huizinga, J. Tang, A. Ecoffet, B. Houghton, R. Sampedro, and J. Clune, "Video pretraining (vpt): Learning to act by watching unlabeled online videos," *Advances in Neural Information Processing Systems*, vol. 35, pp. 24 639–24 654, 2022.
- [24] G. Pavlakos, D. Shan, I. Radosavovic, A. Kanazawa, D. Fouhey, and J. Malik, "Reconstructing hands in 3d with transformers," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 9826–9836.
- [25] M. Lepert, J. Fang, and J. Bohg, "Phantom: Training robots without robots using only human videos," *arXiv preprint arXiv:2503.00779*, 2025.
- [26] —, "Masquerade: Learning from in-the-wild human videos using data-editing," *arXiv preprint arXiv:2508.09976*, 2025.
- [27] S. Zhou, Y. Du, J. Chen, Y. Li, D.-Y. Yeung, and C. Gan, "Robodreamer: Learning compositional world models for robot imagination," *arXiv preprint arXiv:2404.12377*, 2024.
- [28] S. Nair, A. Rajeswaran, V. Kumar, C. Finn, and A. Gupta, "R3m: A universal visual representation for robot manipulation," *arXiv preprint arXiv:2203.12601*, 2022.
- [29] Y. J. Ma, S. Sodhani, D. Jayaraman, O. Bastani, V. Kumar, and A. Zhang, "Vip: Towards universal visual reward and representation via value-implicit pre-training," *arXiv preprint arXiv:2210.00030*, 2022.
- [30] S. Ye, J. Jang, B. Jeon, S. Joo, J. Yang, B. Peng, A. Mandlekar, R. Tan, Y.-W. Chao, B. Y. Lin, *et al.*, "Latent action pretraining from videos," *arXiv preprint arXiv:2410.11758*, 2024.
- [31] R. Punamiya, D. Patel, P. Aphiwetsa, P. Kuppli, L. Y. Zhu, S. Kareer, J. Hoffman, and D. Xu, "Egobridge: Domain adaptation for generalizable imitation from egocentric human data," in *Human to Robot: Workshop on Sensorizing, Modeling, and Learning from Humans*, 2025.
- [32] Y. Du, S. Yang, B. Dai, H. Dai, O. Nachum, J. Tenenbaum, D. Schuurmans, and P. Abbeel, "Learning universal policies via text-guided video generation," *Advances in neural information processing systems*, vol. 36, pp. 9156–9172, 2023.
- [33] S. Kareer, K. Pertsch, J. Darpinian, J. Hoffman, D. Xu, S. Levine, C. Finn, and S. Nair, "Emergence of human to robot transfer in vision-language-action models," *arXiv preprint arXiv:2512.22414*, 2025.
- [34] Z. Li, Y. Yang, S. Xie, K. Xu, T. Chen, Y. Wang, Z. Shen, Y. Shen, Y. Chen, W. Li, Y. Zheng, C. Zhang, S. Lin, F. Teng, H. Yang, M. Chen, S. Xie, and R. Wu, "Lehome: A simulation environment for deformable object manipulation in household scenarios," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2026.
- [35] World Labs, "Marble: A Multimodal World Model," 2025, accessed: 2026-02. [Online]. Available: <https://www.worldlabs.ai/blog/marble-world-model>
- [36] X. Chen, F.-J. Chu, P. Gleize, K. J. Liang, A. Sax, H. Tang, W. Wang, M. Guo, T. Hardin, X. Li, *et al.*, "Sam 3d: 3dfy anything in images," *arXiv preprint arXiv:2511.16624*, 2025.

- [37] B. Wen, W. Yang, J. Kautz, and S. Birchfield, "Foundationpose: Unified 6d pose estimation and tracking of novel objects," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2024, pp. 17 868–17 879.
- [38] Y. Labbé, L. Manuelli, A. Mousavian, S. Tyree, S. Birchfield, J. Tremblay, J. Carpentier, M. Aubry, D. Fox, and J. Sivic, "Megapose: 6d pose estimation of novel objects via render & compare," *arXiv preprint arXiv:2212.06870*, 2022.
- [39] T. Wu, J. Zhang, S. Liang, Z. Han, and H. Dong, "Foundation feature-driven online end-effector pose estimation: A marker-free and learning-free approach," in *2025 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2025, pp. 1921–1928.
- [40] NVIDIA Corporation, "Nvidia isaac sim: High-fidelity simulation for robotics," <https://developer.nvidia.com/isaac-sim>, 2023, accessed: 2026-03-03.